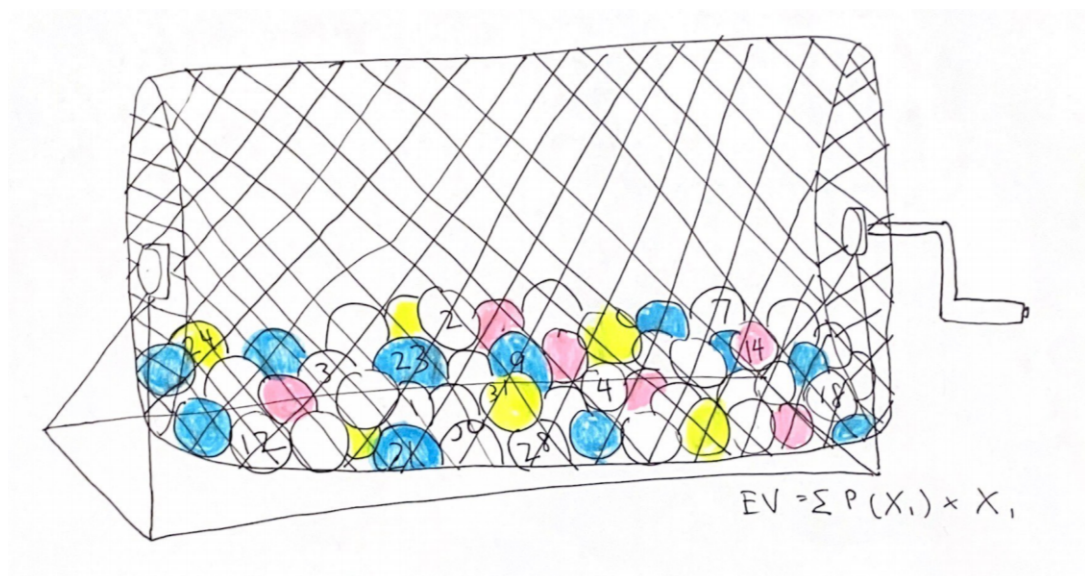


You “MIT” Just Win the Lottery!

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1 Introduction

Those who don't skip Physical Education will know that a recent attendance question asked, “What would you do in life if you knew you could not fail?” Readers of *Math Survey* might be thinking of something nerdy, but there's always the classic, go-to answer of winning the lottery. An old saying calls the lottery a “tax on the stupid,” and many regard it as another form of gambling. It should not be surprising then that the first dated lottery system started with betting on elections in 17th-century Genoa. Yet when there are losers, there must be winners. And with some bugs in the lottery's set-up and some interesting calculations, some may win with chance instead of by chance.

2 A Little History

Genoa chose their two *governatori* not by democracy, not by appointment, and not by nepotism – presumably so. They simply drew the names of two people from a heap of 120 slips of paper with the names of 120 councilors. Personally, it was not long after March 15, 2020 before I started betting donuts on the duration of quarantine, so with the first casinos and gambling houses taking over Italy in the 17th century,

it's reasonable that bored Genoan gamblers saw an opportunity in their electoral system. They put a number to each councilor, and eliminated the need to even show up at pubs on Election Day. Sound familiar?

Adam Smith wrote in *The Wealth of Nations* that “there is not, however, a more certain proposition in mathematics, than that the more tickets you adventure upon, the more likely you are to be a loser. Adventure upon all the tickets in the lottery, and you lose for certain.” Hopefully, most of us are familiar with being told that the more you gamble the more you lose. While this may be good advice for my wallet before I have to take everyone out for a Dunkin’ Donuts buffet, the first part of his argument is not necessarily true. One can argue that buying more tickets for the lottery increases one’s chances of winning. Another can argue that buying more tickets only means spending more money. To put these arguments to numbers, you need something called an *expected value*.

3 A Donut Lottery Example

Let’s say there are 20 different flavored donuts to give away. If I happen to give out my calculus teacher’s mystery favorite flavor, he’d buy me 17 donuts in return (not that he would ever dabble in transactions like this). If I give out all 20 donuts, then I surely would make a whopping profit of negative three donuts. This is what Smith meant by “lose for certain.” In comparison, if I only give out one donut, I’d at least have a $\frac{1}{20}$ chance of winning a profit. If I give out an additional one, I’d have a $\frac{2}{20}$ chance of winning. The breaking point is when I give out 17 donuts. The chance of being rewarded by my teacher or basically an all-for-nothing is $\frac{17}{20}$, and there’s a $\frac{3}{20}$ chance that I lose donuts. Until now, I’ve continuously made myself less likely to lose by giving out more. However, the moment I spend another donut, I’m sure to lose despite my teacher’s favorite flavor. Another thing to consider is that giving out 17 donuts makes me less likely to lose, but the consequences of losing become much more grave than giving out one. Expected value assigns a value to the number of donuts given out with the outcome and the chance of the outcome taken into account. Thus, we calculate it by multiplying each possible outcome by the probability of it happening, and summing up all of those products.

$$EV = \sum P(X_i) \times X_i$$

If we get the right flavor: $\frac{1}{20} \times 17 \text{ donuts} = \frac{17}{20} \text{ donuts}$.

If we don’t: $\frac{19}{20} \times 0 \text{ donuts} = 0 \text{ donuts}$.

Expected value of each donut: $\frac{17}{20} \text{ donuts}$.

By participating, I lowered the value of each donut to that of $\frac{17}{20}$ th of a donut. Maybe I should look for donut flavors that are statistically more appealing to math teachers instead.

Now, most lotteries have way more outcomes other than completely winning or losing, in order to keep people going for the smaller prizes. Most also take cash over donuts. Don’t confuse expected value with unit price either. There is no way anybody would measure out $\frac{17}{20}$ th of a donut just for you. Some other things to take into account are that when prizes go unclaimed, more people enter the prize pool, and the jackpot grows. With this, expected value grows as well. Let’s say that many more people are suddenly interested in betting donuts. My teacher is now willing to give out different prizes for different favorite flavors. There’s a:

- $\frac{1}{3300}$ chance of winning the jackpot of x donuts

- $\frac{1}{2500}$ chance of winning 500 donuts
- $\frac{1}{2000}$ chance of winning 400 donuts
- $\frac{1}{1500}$ chance of winning 250 donuts
- $\frac{1}{1250}$ chance of winning 175 donuts
- $\frac{1}{1000}$ chance of winning 17 donuts like the good old days.

Then the expected value of each donut is now: $\frac{x}{3300} + \frac{500}{2500} + \frac{400}{2000} + \frac{250}{1500} + \frac{175}{1250} + \frac{17}{1000}$ and x changes based on the number of participants.

Now we set this greater than or equal to the one donut you spent:

$$\begin{aligned} \frac{x}{3300} + \frac{500}{2500} + \frac{400}{2000} + \frac{250}{1500} + \frac{175}{1250} + \frac{17}{1000} &\geq 1 \\ \frac{x}{3300} + 0.723667 &\geq 1 \\ \frac{x}{3300} &\geq 0.276333 \\ x &\geq 911.9 \end{aligned}$$

And with some simple algebra, you can find the optimal time to invest in an underground donut. In this example, the expected return value is positive once the jackpot exceeds 911 donuts.

4 The Cash WinFall Example

At this point those here to get information on a good college might be wondering if the title is pure clickbait, but in 2005 an MIT student named James Harvey also realized he could do some quick calculations with Massachusetts' Cash WinFall. The rules of the game were:

- Players have six numbers from 1-46.
- You can win by matching with at least two winning numbers drawn every two weeks.
- Getting all six numbers wins you the jackpot.
- The jackpot starts at \$500,000 and goes up every time there's no winner until it reaches \$2,300,000.
 - Once the jackpot exceeds the maximum value, it resets to \$500,000 and the rest of the money gets distributed among the smaller prizes.

Starting from the jackpot, there's a $\left(\frac{1}{46}\right) = \frac{1}{9366819}$ chance of winning. On a normal day with a million dollar jackpot, the expected value of the ticket could've been \$79.80. With the ticket priced at \$2, this wasn't the best trade. On a rolldown day, \$600,000 was distributed among the 5 and 3 number prizes, while \$1.4 million got put into the 4 number prize. Doing our usual calculations, you'd get that now the ticket's expected value is \$5.53, yielding a 167.5% return. Knowing this, students at MIT bought thousands of tickets and soon, tens of thousands. Not long after, the project turned into a thriving business named Random Strategies. The lottery would close seven years later, but the business made \$3.5 million off of this game.

Random Strategies wasn't the first to discover this, nor Doctor Zhang Lottery Club, nor retiree Gerald Selbee. Eighteenth-century France tried all sorts of ways to drive the economy. An awkward time passed by when John Law tried to introduce a paper money system (1716-1720). Nevertheless, under Louis XV, minister Cardinal Fleury stabilized France's monetary system by implementing a conversion between silver, gold, and coins used across the country. Michel Le Peletier des Forts, a deputy officer, wanted to drive the sale of bonds that funded the government. He attached the right to one ticket with a 500,000-livre prize to each bond sold. Unfortunately, Le Peletier might've skipped a few math classes, and made some *silly mistakes* when designing the lottery. Like Cash WinFall, it had rolldown days. Like Cash WinFall, tickets had a great positive return value on these rolldown days. Like Cash WinFall, buying enough tickets would give you a bigger chance of winning quite a sum. Mathematician Charles-Marie de La Condamine saw this even a couple hundred years ago and gathered his buddies. Among them was writer and public activist Voltaire, who actually gained most of his wealth from investing in this lottery system instead of his writings.

It is legally required for meddlers in the lotto to be at least 18 years old in New York, so before you get excited, know that not all lotteries have obvious loopholes, and that I do not endorse getting fake IDs. However, if in a couple years you do find potential in a certain -Fall or -Ball or -Millions, do ring me up.

5 Problems

- 1) The spinning donut of fortune has five sections. Landing on the blue section gives you \$1000, yellow gives you \$500, white gives you \$100, red gives you \$50, and green gives you another spin. $\frac{1}{50}$ of the donut is blue, $\frac{1}{25}$ of it is yellow, $\frac{1}{20}$ is white, $\frac{1}{10}$ is red, and the rest is green. What is the expected value of each spin?
- 2) We play a game. The pot starts at \$0. On every turn, you flip a fair coin. If you flip heads, I add \$100 to the pot. If you flip tails, I take all of the money out of the pot, and you are assessed a "strike." You can stop the game before any flip and collect the contents of the pot, but if you get three strikes, the game is over, and you win nothing. Find, with proof, the expected value of your winnings if you follow an optimal strategy. (USAMTS Year 17, Round 4, Problem 3)
- 3) Bobby Boo is giving out 100 different donuts. There are 95 different non-chocolate flavored donuts and five chocolate donuts. The donuts all look about the same, and Bobby Boo wants to keep one chocolate donut for himself, so he will stop giving out donuts once he gives out the fourth donut. What is the expected average number of donuts Bobby Boo will give out?
- 4) Blubby Blo put an ant on a 5 x 5 square, and it took four steps (each step crossing to a new square) in a random direction (except back to the square it came from). If the ant ends up in a corner, Blubby Blo wins 10 donuts. If it ends up on an edge, he wins five donuts. Otherwise, he gets nothing. What is the expected number of donuts Blubby Blo will win?
- 5) Crabby Cat has four blue socks, four white socks, four pink socks, and four leopard-print socks. It's the morning, and Crabby Cat is too lazy to look inside her drawer, so she'll just keep pulling out socks until she has two pairs of matching socks. What is the expected average number of socks Crabby Cat will pull out?
- 6) Dooby Doo is taking cookies out of the oven without replacement (because he's eating them). There are five cookies in total, and they are numbered from 1 to 5. He'll stop taking cookies out once he takes out a number greater than the one he just ate. What is the expected number of cookies Dooby Doo eats?

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